
Stochastic Lifting for Generating Trajectories of Stochastic Physical Systems

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Abstract

Many stochastic physical systems evolve smoothly over time in the sense that the distribution of states changes regularly with time. The precise transition from current to next state is often modeled as the interplay of a smooth map and an explicit source of randomness. Stochastic Lifting leverages this premise by attaching an independent, high-dimensional random label to each state transition in the training data and fitting a transition map from the current state and label to the next state using a standard regression loss. The labels act as auxiliary coordinates that let the model represent multiple plausible outcomes for similar current states, avoiding collapse to a mean prediction in the finite-sample size regime. At inference, drawing fresh labels and rolling the map forward generates diverse trajectories with a single network evaluation per time step, with the smoothness bias of the learned map supporting accurate sampling in practice.

1. Overview

Many problems in computational physics require drawing large numbers of trajectories from stochastic time-dependent systems. A common abstraction is that dynamics induce a transition law $\mathbf{X}_{t+1} \sim \rho(\cdot | \mathbf{X}_t = \mathbf{x}_t)$ conditional on the previous state $\mathbf{x}_t$. The learning problem that we study is therefore not point prediction but sampling plausible next states given a current state $\mathbf{x}_t$.

A fundamental difficulty is that fitting a map to training data $(\mathbf{x}_t, \mathbf{x}_{t+1})$ with standard mean-squared-error regression collapses to the conditional expectation $\mathbb{E}[\mathbf{X}_{t+1} | \mathbf{X}_t = \mathbf{x}_t]$ and therefore suppresses the intrinsic variability of the stochastic trajectories. At the other extreme, general-purpose generative models such as diffusion- and flow-based approaches can represent distributions, but they typically require multi-step sampling at inference: roughly speaking, generating

T time steps with an s -step sampler requires $T \times s$ neural-network evaluations. Recent work on one-step samplers aim to reduce this cost by distilling multi-step generation into a single evaluation, but such methods are often more difficult to train reliably and have primarily been demonstrated in stationary settings that differ from the trajectory generation setting that we are interested in. We defer a detailed discussion to the literature review in the next section.

A smoothness premise specific to time-dependent systems and trajectory data Our approach is based on a premise that is natural for physical systems; and more broadly for many time-dependent dynamical systems: for sufficiently small time steps, the conditional transition law $\rho(\cdot | \mathbf{x}_t)$ varies regularly with the state $\mathbf{x}_t$, and next states can be interpreted as small perturbations of the current state. This regularity suggests that the transition can be (at least approximately) represented by a smooth map that takes the current state together with an explicit source of randomness and outputs a next state. The key point is the smoothness of the map. We therefore use smoothness as a guiding heuristic: when a learned transition map interpolates the training data and is smooth, then evaluating it with fresh randomness is expected to generate plausible next-state samples that are close to the desired conditional law.

Stochastic Lifting Stochastic Lifting augments the training-data pairs $(\mathbf{x}_t, \mathbf{x}_{t+1})$ with an independently drawn auxiliary stochastic label ξ_t . It then trains a map $F : (\mathbf{x}_t, \xi_t) \mapsto \mathbf{x}_{t+1}$ by minimizing a standard empirical regression loss over the labeled pairs. The label acts as an auxiliary coordinate that parametrizes stochasticity: by assigning a distinct random label to each realized transition in the training data, the map F can fit multiple possible next states associated with similar or identical current states, rather than being driven by the regression loss to collapse toward the average. Moreover, the dimension of the label provides a handle on regularity to some extent: increasing the label dimension improves the separability of labeled training-data pairs and, as we show, can reduce the minimal Lipschitz constant among interpolants of the training data. In this way, stochastic labeling supports learning maps that interpolate the training data and remain smooth, aligning the learned maps with the smoothness premised outlined above.

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At inference time, we take the current state, draw a fresh label from the label distribution and then evaluate the learned map to obtain a next-state sample. Repeating this procedure with fresh labels yields multiple plausible next-state samples. Furthermore, auto-regressively rolling out the map with a fresh label at each time step leads to a trajectory, requiring only one neural-network evaluation per time step.

Regime and scope An important nuance is that Stochastic Lifting is *not* aiming to yield a meaningful population-limit solution. Indeed, even with the labeled data, the population minimizer collapses to the conditional expectation and ignores the labels. Stochastic Lifting is instead intended for the finite-sample regime, where the labeled data can be fit closely, often up to interpolation. Accordingly, instead of convergence in the classical population limit, we are interested in how close the samples that are generated with Stochastic Lifting are compared to actual samples. Our theory provides some insights into this, most notably showing again that smoothness is the key condition and relating smoothness to the label dimension.

Empirically, we demonstrate state-of-the-art accuracy for generating trajectories of stochastic time-dependent physical systems compared to, e.g., auto-regressive diffusion models and one-step samplers. At the same time, we stress that our method is not designed for unconditional sample generation from noise (e.g., images from noise), where the corresponding map is often globally complex and generally falls outside the regularity regime we rely on. The transition-learning setting, by contrast, provides the structural regularity that makes Stochastic Lifting effective.

Summary of contributions (1) Introduce Stochastic Lifting, which assigns stochastic labels to transitions to learn a next-step generator; sampling fresh labels at inference time enables trajectory generation with a single network evaluation per time step.

(2) Show that smoothness is the key condition that makes next-step sampling from an interpolating transition map both accurate and computationally efficient.

(3) Relate the dimension of the stochastic label to the regularity (Lipschitz) of the transition map, providing a mechanism for promoting smoothness.

(4) Demonstrate state-of-the-art trajectory generation for stochastic, time-dependent physical systems.

2. Setup and problem formulation

2.1. Setup

Consider a stochastic process $\{X_t\}_t \subset \mathcal{X} = [0, 1]^n$ with initial condition $X_0 \sim \rho_0$. We denote a realization of X_t

as x_t and often refer to it as state. The samples evolve via a conditional distribution

$$X_{t+1} | X_t = x_t \sim \rho(\cdot | x_t). \quad (1)$$

To ease exposition, we focus on transitions that condition on the previous state $X_t = x_t$ only, but all of the following extends to transitions that condition on a history of more than one previous state. By the law of total probability, we define the time marginals as $\rho_{t+1} = \int_{\mathcal{X}} \rho(\cdot | x_t) \rho_t(x_t) dx_t$ with the assumption that we can directly sample from ρ_0 . Notice that the time marginals ρ_t can change with time while the conditional distribution $\rho(\cdot | x_t)$ is independent of time and only depends on the previous realization x_t . The coupling between x_t, x_{t+1} is described by the density π_t over $\mathbb{R}^n \times \mathbb{R}^n$ as $\pi_t(x_t, x_{t+1}) = \rho(x_{t+1} | x_t) \rho_t(x_t)$.

Our data consists of pairs of realizations

$$\mathcal{D} = \bigcup_{i=1}^M \bigcup_{t=0}^{T-1} \{(x_t^i, x_{t+1}^i)\} \quad (2)$$

with $(x_t^i, x_{t+1}^i) \sim \pi_t$ for all $i = 1, \dots, M$. The goal is to learn a map F that mimics the transition (1). Once we have such a map, we can roll it out autoregressively to rapidly predict new trajectories.

2.2. Challenges of learning one-step transition functions of stochastic processes

Stochastic dynamics lead to one-to-many maps There cannot exist a function $f : \mathcal{X} \rightarrow \mathcal{X}, x_t \mapsto x_{t+1}$ that describes the transition (1) of the stochastic process $\{X_t\}_t$ over more than one time step. The reason is that, when $\rho(\cdot | x_t)$ does not collapse to a single point, the transition from x_t to $X_{t+1} | X_t = x_t$ can be interpreted as a one-to-many map, see Figure 1a. There are arbitrarily many different realizations $x_{t+1}^j, j = 1, 2, 3, \dots$ of $X_{t+1} | X_t = x_t$ at time $t + 1$ and a function f can map from x_t to only one of them. Even if we consider two different but very close x_t and x'_t , a similar issue can arise when the corresponding x_{t+1} and x'_{t+1} are far apart. One can interpret this as a poorly conditioned regression problem, which requires a regression function (and a parametrization that can represent it) that is far from smooth. In particular, the Lipschitz constant of f grows to infeasible values in this setting.

One interpretation of the one-to-many map perspective is that the transition from x_t to a realization of $X_{t+1} | X_t = x_t$ depends on additional information that is not present in x_t . For example, let us consider a transition (1) stemming from a discretized stochastic differential equation $X_{t+1} = b(X_t) + \sigma W_t$. The transition from X_t to X_{t+1} depends on the realization of X_t but additionally on the realization of the noise W_t , which is typically independent of X_t . Thus, x_t alone is insufficient to describe $X_{t+1} | X_t = x_t$.

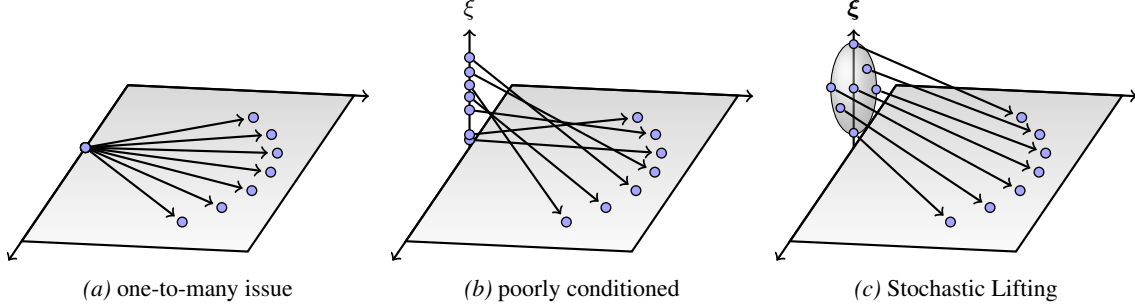


Figure 1. (a) For data from a stochastic process, (almost) identical realizations can have different next realizations, which prevents learning a transition map from just trajectories. (b) Labeling data points separates them but low-dimensional labels can still lead to poorly conditioned regression problems. (c) Stochastic Lifting uses high-dimensional labels, which enables fitting a smooth regression function.

Fitting a function to stochastic dynamics collapses to the mean dynamics Simply fitting a function $f : \mathcal{X} \rightarrow \mathcal{X}$, $x_t \mapsto x_{t+1}$ to data (2) from a stochastic process can capture only mean-like behavior. In fact, training f on data (2) with the mean-squared error loss collapses to the conditional expectation $x_t \mapsto \mathbb{E}[X_{t+1} | X_t = x_t]$ in the limit $M \rightarrow \infty$.

If the stochasticity in the data (2) is due to noise and perturbations, then often the goal is recovering the mean behavior, in which case fitting time integrators with the mean-squared loss is reasonable; as in operator learning (Li et al., 2020; Lu et al., 2021; Kovachki et al., 2023). However, our goal is generating new sample trajectories where the randomness is not only injected at the initial condition sampled from ρ_0 but also during the roll out, as described by the transition (1) and reflected in the data (2) (recall the example with the stochastic differential equation in the previous paragraph). Thus, capturing mean-like behavior only by applying operator learning and other deterministic transition modeling methods and relying on input randomness alone is insufficient. In particular, we will also consider stochastic processes that always start with the same initial condition $\rho_0 = \delta_{x_0}$ (see wave and multi-phase flow in Section 5).

2.3. Literature review

Multi- and single-step generative models There is a range of multi-step methods for generative modeling, which build on diffusion- and flow-based modeling (Sohl-Dickstein et al., 2015; Ho et al., 2020; Hyvärinen, 2005; Song et al., 2019; Song & Ermon, 2019; Song et al., 2021; Albergo & Vanden-Eijnden, 2023; Albergo et al., 2023; Lipman et al., 2022; Liu et al., 2022). In contrast, the focus of our work lies on a one-step method, for which a single neural-network function evaluation per realization is sufficient. Early one-step methods are difficult to train (Goodfellow et al., 2014; Arjovsky et al., 2017) or easily collapse (Kingma et al., 2013). There are distillation approaches for constructing one-step models out of multi-step models but these require

having readily available a multi-step model (Yin et al., 2024; Sauer et al., 2024; Zhou et al., 2024; Song et al., 2023; Geng et al., 2024; Frans et al., 2024). More recently, there are methods that aim to circumvent having to train a multi-step model first (Song et al., 2023; Zhou et al., 2025; Geng et al., 2025; Boffi et al., 2025) but they focus on static generation tasks such as noise to image and do not explicitly leverage the regularity induced by conditioning on previous states in time-dependent problems that we rely on for generating trajectory data.

Autoregressive generative models for trajectory data generation There is a range of works on autoregressive diffusion models (ARDMs) (Hoogeboom et al., 2021; Albergo et al., 2023; Price et al., 2025) and also flow-based models (Davtyan et al., 2023; Chen et al., 2024) that can generate trajectory data; however, these require multiple steps per realization, which can be expensive. There are also time-space approaches (Ho et al., 2022) with the drawback of having to handle a large time-space tensor. We also mention marginal trajectory matching (Neklyudov et al., 2023; Berman et al., 2024; Blickhan et al., 2025) which also performs only a single neural-network function evaluation per time step but the corresponding models can be challenging to scale to high dimensions. Another line of work imposes constraints on the diffusion trajectories to better align them (Rühling Cachay et al., 2023; Ruhe et al., 2024); these again require multiple steps per realization.

Lifting A key aspect of our approach is lifting data into higher dimensions, which is prevalent in machine learning (Cortes & Vapnik, 1995; Rahimi & Recht, 2007) but also in computational science in general (McCormick, 1976; Gu, 2011; Kramer & Willcox, 2019; Qian et al., 2020; McQuarrie et al., 2021). We mention neural ordinary differential equations (Dupont et al., 2019; Kidger, 2022) that augment the state space to extend the expressivity of neural ordinary differential equations. With lifting we pursue an analogous goal of making a problem better behaved, namely smoother.

3. Stochastic Lifting

3.1. Labeling data points to avoid one-to-many issue

We randomly draw labels $\xi_t^i \in \mathbb{R}^d$ from a distribution ν for all x_t^i in the data set $\mathcal{D}$, which leads to the augmented data set

$$\mathcal{D}_\xi = \bigcup_{i=1}^M \bigcup_{t=0}^{T-1} \{(x_t^i, x_{t+1}^i, \xi_t^i)\}. \quad (3)$$

In all of the following, the label distribution ν will be $\mathcal{N}(0, I_d)$, the standard normal distribution of dimension d , which implies that the labels are unique almost surely. We assume that ν is independent of t , but we still denote $\xi_t \sim \nu$ to emphasize that, for a given trajectory $x_0^i, \dots, x_{T-1}^i$, the label of the realization at time t need not be the same as the label of the realization at time $t+1$.

Uniquely labeling the data points overcomes the one-to-many issue: Even if there are $x_t^i = x_t^j$ for $i \neq j$, the corresponding labeled points (x_t^i, ξ_t^i) and (x_t^j, ξ_t^j) are unique; see Figure 1b. Thus, circling back to the challenges described in Section 2.2, because of the unique label for each data point, there now exists a function $F : \mathcal{X} \times \mathbb{R}^d \rightarrow \mathcal{X}$ that can interpolate (memorize) the data in the sense

$$F(x_t^i, \xi_t^i) = x_{t+1}^i, \quad (4)$$

for $i = 1, \dots, M$ and $t = 0, \dots, T-1$. Recalling the example from Section 2.2 about the stochastic differential equation, we can interpret ξ_t^i as a surrogate of the realization of noise dW_t that enters the dynamics.

3.2. Wasserstein-2 bound for smooth interpolator

We now show that if the map F is regular in terms of its Lipschitz constant and interpolates the data as in (4), then samples generated with F are close to ρ_{t+1} in the Wasserstein-2 metric.

Proposition 3.1. *Let F interpolate the training data $\mathcal{D}_\xi$ as in (4). At any time t , consider $\tilde{M}$ test samples $\mathcal{D}_\xi^{\text{test}} = \{(\tilde{x}_t^i, \tilde{x}_{t+1}^i, \tilde{\xi}_t^i)\}_{i=1}^{\tilde{M}}$ sampled from $\pi_t \otimes \nu$. Evaluate now the map F to obtain the generated samples $\hat{x}_{t+1}^i = F(\tilde{x}_t^i, \tilde{\xi}_t^i)$ for $i = 1, \dots, \tilde{M}$. Then,*

$$\begin{aligned} \mathbb{E}_{\mathcal{D}_\xi, \mathcal{D}_\xi^{\text{test}}} [W_2(\hat{\rho}_{t+1}, \tilde{\rho}_{t+1})^2] \\ \leq C(1 + 2L_F^2) \min(M, \tilde{M})^{-2/\alpha}, \end{aligned} \quad (5)$$

where $\hat{\rho}_{t+1}$ and $\tilde{\rho}_{t+1}$ are the empirical measures corresponding to the generated samples $\{\hat{x}_{t+1}^i\}_{i=1}^{\tilde{M}}$ and the test samples $\{\tilde{x}_{t+1}^i\}_{i=1}^{\tilde{M}}$, $C > 0$ is a constant, L_F is the Lipschitz constant of F , and $\alpha \geq 0$ depends on the intrinsic dimensions of ρ_t, ρ_{t+1} and ν , whichever is largest.

A proof can be found in Appendix B.2. The bound (5) critically depends on the smoothness of F in terms of the

Lipschitz constant L_F . While simply interpolating independently drawn samples from two distributions can lead to regression functions with high Lipschitz constants in low dimensions (see Figure 1b), we will discuss in the following that labels of high dimension can help to reduce L_F .

Furthermore, a strong conditioning induced by the previous state can be helpful. In particular, even stochastic physics systems lead to trajectories that gradually change over time in the sense that $\|x_{t+1}^i - x_t^i\| = \mathcal{O}(dt)$ scales with a small time-step size dt for all times and realizations. The following corollary shows that if the next realization depends on the current realization and a stochastic update term $R(x, \xi)$ scaled by the time-step size dt , then the scaling dt enters in front of the Lipschitz constant to help alleviate non-smoothness.

Corollary 3.2. *If $F(x, \xi) = x + dtR(x, \xi)$, then (5) can be written as*

$$\begin{aligned} \mathbb{E}_{\mathcal{D}_\xi, \mathcal{D}_\xi^{\text{test}}} [W_2(\hat{\rho}_{t+1}, \tilde{\rho}_{t+1})^2] \\ \leq C(3 + 4dt^2 L_R^2) \min(M, \tilde{M})^{-2/\alpha}, \end{aligned} \quad (6)$$

where C and α are as in Proposition 3.1 and L_R is the Lipschitz constant of R .

Similar bounds can be derived when assuming more structure on F such as $F(x, \xi) = x + dtb(x) + \varepsilon\sigma(\xi)$; see Appendix B.2 and compare to the discrete stochastic differential equation in Section 2.2.

3.3. Lifting for smoothness

The key in the previous results was the smoothness of F given by the Lipschitz constant, which we can control to some extent by using high-dimensional labels. The minimal Lipschitz constant of a function F that interpolates the data as in (4) is

$$L(\mathcal{D}_\xi) = \max_{i \neq j, t} \frac{\|x_{t+1}^i - x_{t+1}^j\|_2}{\sqrt{\|x_t^i - x_t^j\|_2^2 + \|\xi_t^i - \xi_t^j\|_2^2}}, \quad (7)$$

for $i, j = 1, \dots, M$ and $t = 0, \dots, T-1$. Note that a function interpolating the data (4) with Lipschitz constant (7) on the data can be extended to $\mathcal{X} \times \mathbb{R}^d$ without increasing the Lipschitz constant (Federer, 1996)[Theorem 2.10.43]. In practice, we rely on standard deep learning regularization techniques (weight decay, normalization layers) to learn a regular interpolant of the data.

High-dimensional labels decrease the minimal Lipschitz constant

We draw our labels from a standard normal of dimension d . For the sake of the theoretical argument we consider normalized labels with unit norm $\|\xi_t^i\|_2 = 1$, which means that $\frac{1}{2}\|\xi_t^i - \xi_t^j\|_2^2 = 1 - \xi_t^i \cdot \xi_t^j$. That is, the distance

between labels is controlled by their inner product. Because the normalized labels are uniformly distributed on the unit sphere $\mathbb{S}^{d-1}$, $\xi_t^i \cdot \xi_t^j$ is of order $1/\sqrt{d}$ with high probability (Vershynin, 2018) [Section 3.3.3 and Theorem 3.3.9]. Hence increasing d widens the separation between labels and reduces the minimal Lipschitz constant of any interpolant as the following proposition shows (see also Figure 1c).

Proposition 3.3. *For normalized data and normalized labels, if $M \geq 2$ and $d \gtrsim \min\{c_\delta^2 \ln((T+1)M), 2\}$, then*

$$L(\mathcal{D}_\xi) \leq \frac{\sqrt{n}}{\sqrt{2}} \left(1 + c_\delta \sqrt{\frac{\ln((T+1)M)}{d}} \right),$$

holds with probability at least $1 - \delta$, $\delta \in (0, 1)$, and with constant $c_\delta \geq 0$ independent of T, M, d .

See Appendix B.1 for a proof. Thus, we have some control over the Lipschitz constant via the dimension d . The regularity of F is further improved by the closeness between x_t and x_{t+1} (Corollary 3.2). We do not normalize our labels in practice because we expect the components of the label ξ_t^i to be of order one, which is true when $\xi_t^i \sim \mathcal{N}(0, \mathbf{I}_d)$. The argument that $L(\mathcal{D}_\xi)$ is improved by large d remains unchanged.

Orthogonal labels avoid inducing structure Traditionally, one would aim to arrange (couple) labels ξ_t^i and targets (next state) x_{t+1}^i . We do not rearrange because of costs, and thus we need to ensure that no spurious structure is induced between labels and next states. Choosing the labels close to orthogonal achieves this.

3.4. Fitting maps to stochastically lifted data and one-step inference

Training on lifted data and one-step inference We now formulate a regression problem using the labeled data $\mathcal{D}_\xi$. Let us parametrize F as a neural-network function F_θ with weights $\theta \in \mathbb{R}^p$.

We then propose to fit F_θ to the labeled data $\mathcal{D}_\xi$ using a regression based loss function,

$$\mathcal{L}(\theta, \mathcal{D}_\xi) = \frac{1}{MT} \sum_{i=1}^M \sum_{t=0}^{T-1} \|F_\theta(x_t^i, \xi_t^i) - x_{t+1}^i\|_2^2. \quad (8)$$

Other regression loss functions can be used. In the physics based experiments we use a L2 error. In the experiments involving natural video we use an LPIPS loss (Zhang et al., 2018); see Appendix G.

To generate a new state at time $t+1$ for a given x_t , we draw a new label $\xi_t \sim \nu$ and evaluate the $F_\theta(x_t, \xi_t) = x_{t+1}$

to obtain x_{t+1} . This is one-step inference because only a single function evaluation is necessary to generate a new sample.

Lifting avoids collapse onto conditional expectation (mean behavior) of empirical-risk minimizer Critically, when taking the infinite data limit $M \rightarrow \infty$, then the minimizer of (8) collapses to the conditional expectation $\mathbb{E}[x_{t+1} | x_t = x_t]$, since the labels ξ_t^i are drawn independently from x_t^i . This stresses that the map F that we learn with Stochastic Lifting does *not* converge in a distributional sense to the conditional law $\rho(\cdot | x_t)$ in the conventional limit of taking the number of data points to infinity, $M \rightarrow \infty$.

However, this collapse is evitable at finite sample size. We argue that if the label dimension d is allowed to scale with the number of data points $T \times M$, and if the network F_θ is sufficiently expressive, then the minimizer of the *empirical* mean-squared error loss can interpolate the training data as in (4) and does not collapse to the conditional expectation. In particular, for finitely many training data points $M < \infty$, we can achieve zero training error (interpolation/memorization) by ensuring that F_θ is expressive enough while at the same time choosing the label dimension d so high that the minimal Lipschitz constant (7) is controlled, in which case we can expect that F_θ is smooth. In this regime, we expect F_θ to behave as discussed in Proposition 3.1, implying that evaluating F_θ at fresh labels yields outputs that approximately follow the conditional distribution $\rho(\cdot | x_t)$. In practice, Adam with weight decay, normalization layers, and standard architecture choices further bias optimization towards learning globally smooth interpolants; see Appendix G.

As a side remark, we note that in the case of linear least-squares regression, it has been shown that when the ratio $d/(M \times (T+1))$ between dimension d and number of data points $M \times (T+1)$ remains fixed in the limit $d, M \rightarrow \infty$, then collapse is avoided (Hastie et al., 2022).

4. Demonstration on an SDE example

We demonstrate Stochastic Lifting on a duffing oscillator with random forcing, which is described by an SDE with two-dimensional states; see Appendix E.1. We generate training data, augment each pair (x_t, x_{t+1}) with a label, and then train F_θ represented as an multi-layer perceptron (MLP) with the loss (8). At inference time, we rollout the learned F_θ as described in above from a new sample of our initial distribution and fresh labels in each time step.

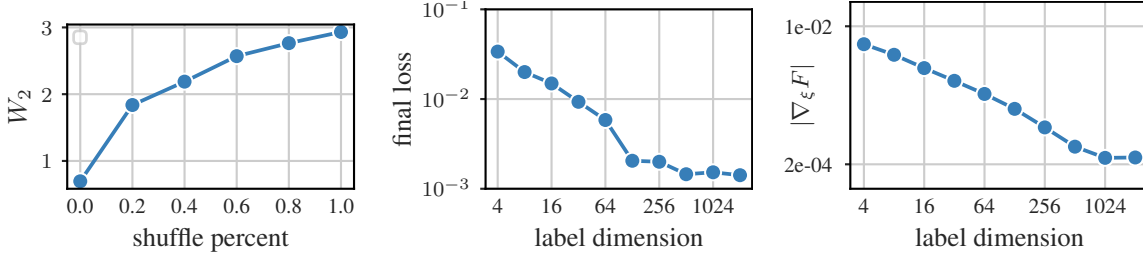


Figure 2. Shuffling training pairs $(\mathbf{x}_t, \mathbf{x}_{t+1})$ breaks the conditional transition structure and degrades Stochastic Lifting’s performance, in agreement with the smoothness premise outlined in the Introduction (left). Increasing the label dimension helps to push the training into the interpolation regime (middle) and induces smoother dependence on the label (right), consistent with our discussion.

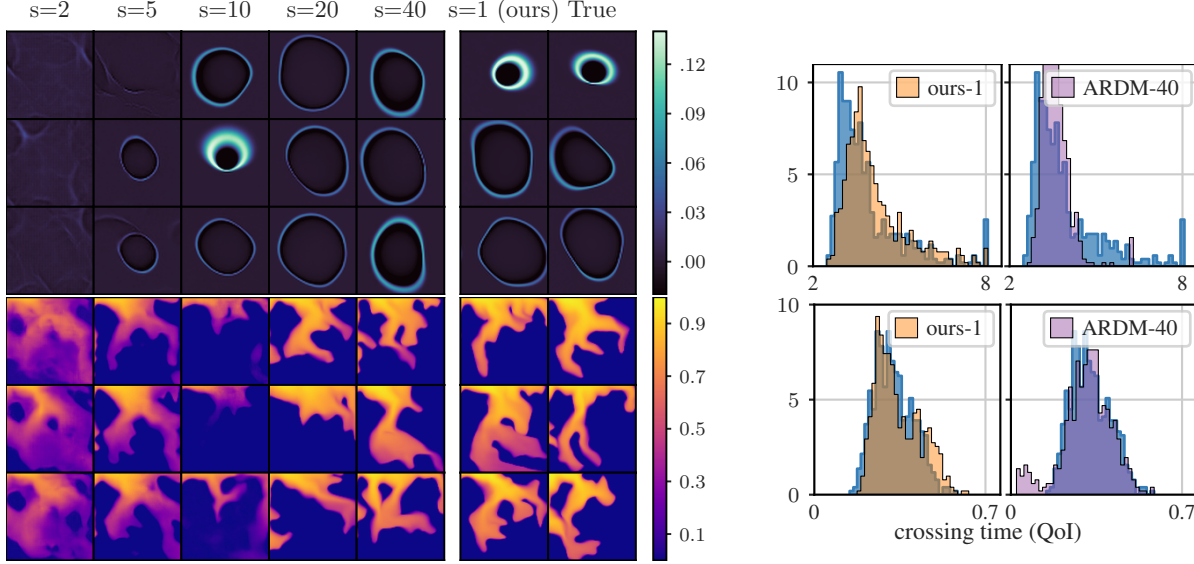


Figure 3. **Left:** Stochastic Lifting generates diverse plausible trajectories in just one step per realization, where ARDMs only begin to generate plausible trajectories when using 20 diffusion steps. **Right:** We compute the crossing time for each trajectory (see appendix E.5) and plot the distribution across 512 generated trajectories vs the ground truth. Stochastic Lifting accurately approximates the ground truth data in distribution, even outperforming ARDM using 40 diffusion steps.

Stochastic Lifting crucially relies on smoothness premise

In Figure 2 (left), we plot W_2 error on test data against the percentage of training pairs $(\mathbf{x}_t, \mathbf{x}_{t+1})$ that are shuffled prior to training. Concretely, we choose a specified percentage of all training trajectories and, at each time step, independently permute the corresponding $\mathbf{x}_{t+1}$ values, thereby breaking the pairing between $\mathbf{x}_t$ and $\mathbf{x}_{t+1}$. As this shuffle percentage increases, we progressively move toward independent samples from the time marginals ρ_t . We observe that increasing the shuffling percentage leads to a larger W_2 error, indicating that Stochastic Lifting indeed critically depends on the smoothness premise in the sense that the conditional transition law $\rho(\cdot|\mathbf{x}_t)$ varies smoothly with $\mathbf{x}_t$ and next states can be interpreted as small perturbations of the current state.

Higher label dimension enables interpolation and promotes smoothness

In Figure 2(middle), we plot the final training loss versus label dimension. As the label dimen-

sion increases, the loss drops to near zero, indicating that higher-dimensional labels enable the network to interpolate the training data. In Figure 2(right), we plot the norm of the network gradient with respect to the label dimension. The gradient norm decreases as the label dimension increases, which indicates that the network becomes smoother in the label, supporting our discussion that higher-dimensional labels can promote smoothness.

5. Numerical experiments

Setup For all of the following experiments we parametrize our map F_θ via UNet architectures (Ronneberger et al., 2015) but our method works with any other standard diffusion backbone (Peebles & Xie, 2023; Bao et al., 2023; Hoogeboom et al., 2023). We repurpose the diffusion-time input to condition on our label ξ_t ; see Appendix F for details.

For all datasets we provide uncured samples from our model in Appendix H.

We stress that we can use Stochastic Lifting directly on the realization pixels and so avoid having to rely on a latent embedding via pre-trained encoder and decoder networks as many other generative modeling methods (Rombach et al., 2022). By avoiding the latent embedding we ease the significant engineering complexity and hyper-parameter tuning that comes with latent embeddings.

With just one step per realization, Stochastic Lifting achieves comparable accuracy as multi-step models on physics problems

We consider two physics problems: first, a traveling wave through a spatially varying random medium, which is motivated by seismic wave propagation (Sato et al., 2012); see Appendix E.2. Second, an incompressible two-phase flow in random porous media, which is motivated by petroleum engineering (oil/water) and models of groundwater flow; see Appendix E.3. Importantly, for both wave and flow problem, the initial condition is deterministic and fixed, thus deterministic approaches such as neural operators cannot capture the stochasticity induced by the random media and permeability fields during the autoregressive roll out; see Section 2.2.

We compare our Stochastic Lifting to an autoregressive diffusion model (ARDM) trained on the same dataset. We use the implementation given in (Kohl et al., 2023), also built around a UNet backbone and comparable parameter count as we use for Stochastic Lifting. We vary the number of diffusion steps s and compare the accuracy to Stochastic Lifting that uses only a single step ($s = 1$).

For the wave and flow problems, ARDM requires 40 steps to generate accurate samples; see Figure 3(left). In contrast, Stochastic Lifting generates accurate samples in one step. Figure 7–8 in the appendix show nearest-neighbor trajectories and realizations and so provide evidence that Stochastic Lifting is indeed generating new samples instead of simply retrieving memorized ones.

To quantitatively compare the generated samples, we consider the crossing time as a physical quantity of interest, which is the time needed that the wave hits the boundary and the saturation reaches the right-bottom corner of the domain, respectively; see Appendix E.5. We plot the distribution of the crossing time of 512 generated trajectories. Stochastic Lifting accurately matches the ground truth data in distribution; see Figure 3(right). Notice that Stochastic Lifting approximates well the non-Gaussian behavior of the crossing time in the wave example (heavy tail), whereas ARDM fails to capture it even with $s = 40$ steps. Table 1 shows the Wasserstein-2 distance between the distribution of the crossing time (WCT) obtained with Stochastic Lifting and ARDM over various steps. Table 1 also plots the

Table 1. Wasserstein distance between the distribution of 512 samples from the true model vs. the generative model measured on various quantities of interest; see Appendix E.5.

	Wave		Flow	
	WCT↓	WIM↓	WCT↓	WIM↓
ARDM $s=2$	2.49e−2	9.11e−5	1.18e−1	3.46e−2
ARDM $s=5$	2.13e−2	1.04e−4	3.39e−2	4.58e−2
ARDM $s=10$	1.04e−2	6.00e−5	9.13e−2	1.29e−1
ARDM $s=20$	7.28e−3	1.06e−4	3.25e−2	1.27e−3
ARDM $s=40$	1.11e−2	9.85e−5	1.86e−3	6.00e−4
SL (ours) $s=1$	4.53e−3	2.06e−5	9.47e−4	6.08e−4

Table 2. Stochastic lifting outperforms all other one-step methods on the BAIR video-generation dataset.

One-step methods ($s = 1$)		
SV2P — (Babaeizadeh et al., 2017)		262.5
SAVP — (Lee et al., 2018)		109.8
DVD-GAN — (Clark et al., 2019)		109.8
TrIVD-GAN-FP — (Luc et al., 2020)		103.3
FitVid — (Babaeizadeh et al., 2021)		93.6
NUWA — (Wu et al., 2022)		86.9
Stochastic Lifting (ours)		69.0
Multi-step methods		
RIVER — (Davtyan et al., 2023)	$s=100$	106.1
MCVD — (Voleti et al., 2022)	$s=1000$	98.8
RaMViD — (Höppe et al., 2022)	$s=750$	84.2
Video Diffusion — (Ho et al., 2022)	$s=16$	66.9
Rolling Diffusion — (Ruhe et al., 2024)	$s=32$	59.6

Wasserstein-2 distance (WIM) for another physics quantity of interest, the integrated mass; see Appendix E.5.

Stochastic Lifting is robust to label dimension, once sufficiently high to allow interpolation In Figure 6(left) we plot the final L2 training loss achieved after optimization. For low label dimensions, the neural network cannot interpolate the data, resulting in high training loss. Figure 6(middle) shows that for a sufficiently high label dimension (in particular once interpolation is achieved), the model produces an accurate approximation in the WTC metric (see Table 1). These results align well with Proposition 3.1. At the same time, the approach is fairly robust to the label dimension once it is sufficiently high because architectures such as UNet and the corresponding modulation (see Appendix F) can compress the labels if necessary.

Stochastic lifting scales to long-time rollouts (CLEVRER) The CLEVRER dataset (Yi et al., 2020) is a synthetic dataset designed specifically for video

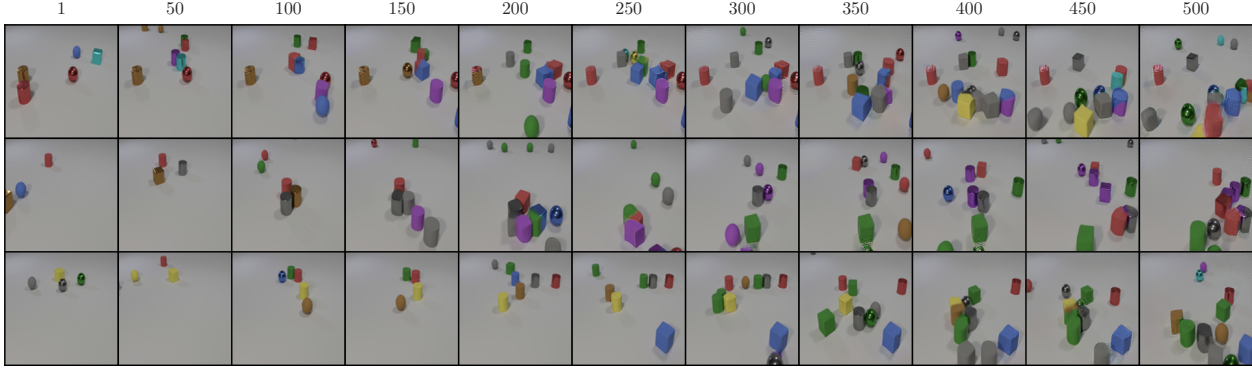


Figure 4. We generate 500 realizations of a 128×128 color video in 0.28 seconds on an H100 GPU. The rollout is stable despite being trained on video with only 16 realizations. In each trajectory, many more objects are in realization than ever occur during training.

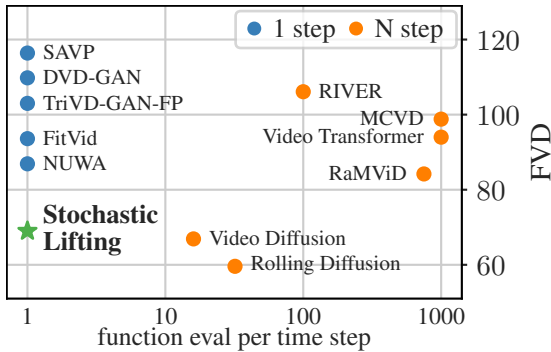


Figure 5. On the BAIR video generation benchmark, Stochastic Lifting is state-of-the-art among one-step methods and competitive with diffusion-based methods that use over $10\times$ more compute.

prediction and reasoning tasks. The videos capture diverse objects moving at high speeds from off realization and colliding with each other. There is inherent stochasticity in when and what objects come into realization. The training videos are 16 realizations long. For testing, we evaluate our model starting with unseen initial realizations and then roll out for 500 realizations, well over one order of magnitude longer than the training videos (compare to 14 realizations in (Chen et al., 2024), 120 realizations in (Davtyan et al., 2023), 64 realizations in (Mei & Patel, 2023)). The long rollout results in objects continuing to enter from off realization and accumulating in clusters as they collide. Importantly, Stochastic Lifting is able to stably generate the interactions and accumulation of objects, even though such accumulations do not occur in the training set.

Stochastic Lifting outperforms other one-step methods on BAIR We apply Stochastic Lifting to the Berkeley AI Research (BAIR) robot pushing dataset, which is a standard benchmark (Ebert et al., 2017). It contains roughly 44000 videos at 64×64 resolution. The standard prediction task is to generate 15 realizations conditioned on one starting realization. We calculate FVD (using the I3D network (Carreira

& Zisserman, 2017)) via the standard procedure on BAIR by generating 100 new videos starting from 256 initial realizations taken from the test set. FVD is then computed between the generated and test videos. Table 2 shows that Stochastic Lifting achieves the lowest (best) FVD (Unterthiner et al., 2018) over current state-of-the-art one-step generation approaches, and starts closing the gap to multi-step methods (Ruhe et al., 2024).

6. Conclusions and limitations

Conclusions We provide evidence with our Stochastic Lifting approach that one can obtain a non-deterministic emulator using a standard empirical regression loss on the data augmented with stochastic labels. We show that interpolation plus smoothness leads to accurate samples in the Wasserstein-2 metric. Empirical results demonstrate that high-dimensional stochastic labels enable generating samples stochastic physics problems and on video generation benchmarks that reproduce statistical properties of the data.

Limitations (a) We critically build on the strong coupling given by the pair of current and next realization in the training data, which means that Stochastic Lifting fails when aiming to generate images from noise. (b) Increasing the label dimension eventually guarantees the existence of a linear interpolant. However, doing so may require proportionally more data; beyond a point, the regression function cannot be trained accurately. (c) Ultimately, the quality of samples created by Stochastic Lifting relies heavily on the fact that the trained network builds a meaningful embedding of the data such that new labels lead to samples within the data manifold. Additional research and theoretical insights are needed to understand when this is the case and when it fails.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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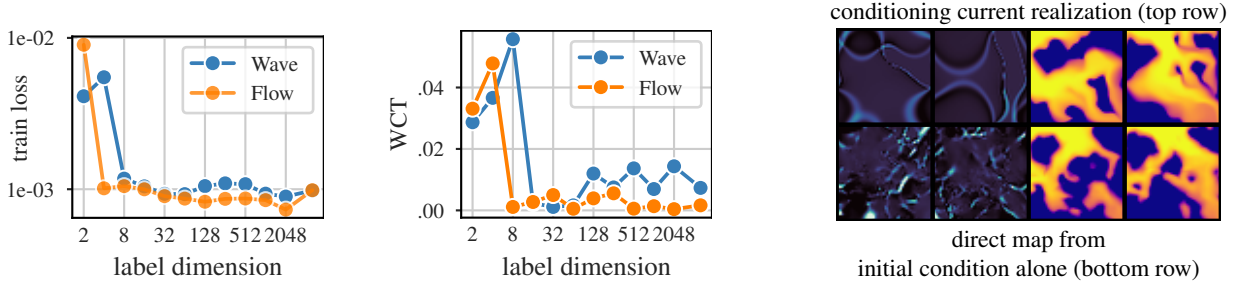


Figure 6. **Left:** Choosing a higher label dimension d allows the neural network to interpolate the data, resulting in one order-of-magnitude lower L2 final training loss. **Middle:** Stochastic Lifting is robust to the choice of the label dimension as long as it is sufficiently high for interpolation. **Right:** Stochastic Lifting critically depends on the conditioning on the current realization to generate a high-quality next realization. If we attempt to jump to the last realization directly, the method breaks down.

A. Additional Results

Stochastic lifting critically depends on conditioning of current realization In Figure 6(right, top) we show the final realization of the rollout from our model which is accurate. In the bottom row, we show what happens when we train a Stochastic Lifting model which maps directly from the initial condition x_0 to the final state x_{T-1} (i.e. without sequential rollout). In this case, Stochastic Lifting breaks, producing near noise in the wave problem and non-physical, disconnected flows in the flow problem. This study provides further evidence that Stochastic Lifting crucially depends on the “closeness” between the distributions corresponding to the current and next realization, which aligns well with Corollary 3.2.

B. Proofs

B.1. Proof of discrete Lipschitz constant bound

Proposition B.1. Assume that all x_t^i are normalized so that $x_t^i \in [0, 1]^n$. Assume further that the labels ξ_t^i are sampled uniformly from the unit sphere $\mathbb{S}^{d-1}$, which corresponds to sampling standard normals and normalizing them afterwards. Assume further that $M \geq 2$ and $d \geq \min\{c_\delta^2 \ln((T+1)M), 2\}$ with

$$c_\delta = \frac{\sqrt{2}}{\sqrt{c}} + \sqrt{\frac{2 \ln(2/\delta)}{c \ln(2)}}, \quad (9)$$

where $c > 0$ is a constant independent of M, d, T . For $\delta \in (0, 1)$, with probability at least $1 - \delta$, it holds

$$L(\mathcal{D}_\xi) \leq \frac{\sqrt{n}}{\sqrt{2}} \left(1 + c_\delta \sqrt{\frac{\ln((T+1)M)}{d}} \right). \quad (10)$$

Proof. First, notice that

$$\Delta_{\max} = \max_{i,j,t} \|x_t^i - x_t^j\|_2 \leq \sqrt{n}$$

because of the normalization $x_t^i \in [0, 1]^n$. For a fixed triplet (i, j, t) , let us consider

$$\frac{\|x_{t+1}^i - x_{t+1}^j\|_2}{\sqrt{\|x_t^i - x_t^j\|_2^2 + \|\xi_t^i - \xi_t^j\|_2^2}} \leq \frac{\sqrt{n}}{\|\xi_t^i - \xi_t^j\|_2},$$

because $\|x_t^i - x_t^j\|_2 \geq 0$ and $\Delta_{\max} \leq \sqrt{n}$. Thus, it suffices to lower bound $\min_{i,j,t} \|\xi_t^i - \xi_t^j\|_2$.

Because the labels ξ_t^i are uniformly sampled from the unit sphere $\mathbb{S}^{d-1}$, for the inner product, we have the following concentration inequality that follows from (Vershynin, 2018, Exercise 5.1.12)

$$\mathbb{P}[|\langle \xi_t^i, \xi_t^j \rangle| \geq \epsilon] \leq 2 \exp(-c d \epsilon^2), \quad \epsilon \in [0, 1] \quad (11)$$

for a universal constant $c \geq 0$. (To see (11): Set $f_u : \mathbb{S}^{d-1} \rightarrow \mathbb{R}, z \mapsto \langle z, u \rangle$ for a fixed $u \in \mathbb{S}^{d-1}$. For any $y, z \in \mathbb{S}^{d-1}$ obtain that $\|f_u(y) - f_u(z)\|_2 = \|\langle y - z, u \rangle\|_2 \leq \|y - z\|_2 \|u\|_2 = \|y - z\|_2$ because $\|u\|_2 = 1$ by definition. Thus the Lipschitz constant of f_u is one. Now notice that $\mathbb{E}[f_u(z)] = 0$ because $\mathbb{E}[z] = 0$ for z uniformly on $\mathbb{S}^{d-1}$. Therefore with (Vershynin, 2018, Exercise 5.1.12) obtain that $\mathbb{P}[|f_u(z)| \geq \epsilon] \leq 2 \exp(-cd \epsilon^2)$ for $\epsilon \in (0, 1]$. Taking two independent ξ_t^i, ξ_t^j uniformly on $\mathbb{S}^{d-1}$, we obtain $\mathbb{P}[\langle \xi_t^i, \xi_t^j \rangle \geq \epsilon] = \mathbb{E}_{\xi_t^i}[\mathbb{P}[\langle \xi_t^i, \xi_t^j \rangle \geq \epsilon | \xi_t^i]] \leq \mathbb{E}_{\xi_t^i}[2 \exp(-cd \epsilon^2)] = 2 \exp(-cd \epsilon^2)$ because the right-hand side of the bound for f_u is independent of ξ_t^j .)

Thus, we obtain

$$\mathbb{P}[\sqrt{2(1 - |\langle \xi_t^i, \xi_t^j \rangle|)} \leq \sqrt{2(1 - \epsilon)}] \leq 2 \exp(-cd \epsilon^2), \quad \epsilon \in [0, 1],$$

holds. This is useful because $\|\xi_t^i - \xi_t^j\|_2^2 = 2 - 2\langle \xi_t^i, \xi_t^j \rangle$ and thus $\|\xi_t^i - \xi_t^j\|_2 \geq \sqrt{2(1 - |\langle \xi_t^i, \xi_t^j \rangle|)}$. Selecting a $\delta \in (0, 1)$ and using a union bound, we obtain

$$\mathbb{P}[\min_{i,j,t} \|\xi_t^i - \xi_t^j\|_2 \geq \sqrt{2(1 - \epsilon_\delta)}] \geq 1 - \delta,$$

for

$$\epsilon_\delta = \sqrt{\frac{1}{cd} (\ln(\hat{M}) + \ln(2/\delta))}$$

where $\hat{M} = (T+1) \binom{M}{2}$.

We have $\ln(\hat{M}) \leq 2 \ln((T+1)M)$ because $T \geq 0$ and under the assumption $M \geq 2$,

$$\begin{aligned} \ln(\hat{M}) &= \ln\left((T+1) \frac{M(M-1)}{2}\right) \\ &= \ln(T+1) + \ln\left(\frac{M(M-1)}{2}\right) \\ &\leq \ln(T+1) + \ln\left(\frac{M^2}{2}\right) \quad (\text{because } M(M-1) \leq M^2) \\ &= \ln(T+1) + 2 \ln M - \ln 2 \\ &\leq \ln(T+1) + 2 \ln M \quad (\text{dropping the negative term } -\ln 2) \\ &= [\ln(T+1) + \ln M] + \ln M \\ &= \ln((T+1)M) + \ln M \\ &\leq 2 \ln((T+1)M) \quad (\text{since } \ln M \leq \ln((T+1)M)), \end{aligned}$$

and thus obtain

$$\epsilon_\delta^2 = \frac{1}{c} \left(\frac{2 \ln((T+1)M)}{d} + \frac{\ln(2/\delta)}{d} \right),$$

which leads to

$$\epsilon_\delta \leq \frac{1}{\sqrt{c}} \left(\sqrt{\frac{2 \ln((T+1)M)}{d}} + \sqrt{\frac{\ln(2/\delta)}{d}} \right)$$

Because $\ln((T+1)M) \geq \ln(2)$ (because we have $(T+1)M \geq 2$),

$$\sqrt{\frac{\ln(2/\delta)}{d}} = \sqrt{\frac{\ln(2/\delta)}{\ln(2)}} \sqrt{\frac{\ln(2)}{d}} \leq \sqrt{\frac{\ln(2/\delta)}{\ln(2)}} \sqrt{\frac{\ln((T+1)M)}{d}}.$$

Now define c_δ as in (9) to obtain

$$\epsilon_\delta \leq c_\delta \sqrt{\frac{\ln((T+1)M)}{d}}.$$

With (11), we have with probability at least $1 - \delta$ that

$$\max_{i,j,t} \frac{\sqrt{n}}{\|\xi_t^i - \xi_t^j\|_2} \leq \frac{\sqrt{n}}{\sqrt{2(1 - \epsilon_\delta)}},$$

which is non-vacuous as long as $d \geq c_\delta^2 \ln((T+1)M)$ so that $\epsilon_\delta \leq 1$. We further have $\sqrt{2(1 - \epsilon_\delta)} \geq \sqrt{2}(1 - \frac{1}{2}\epsilon_\delta)$ and thus

$$\frac{\sqrt{n}}{\sqrt{2(1 - \frac{1}{2}\epsilon_\delta)}} \leq \frac{\sqrt{n}}{\sqrt{2}}(1 + \epsilon_\delta) \leq \frac{\sqrt{n}}{\sqrt{2}} \left(1 + c_\delta \sqrt{\frac{\ln((T+1)M)}{d}} \right),$$

which shows the bound (10). $\square$

B.2. Interpolation of training data + smoothness = bound on test data

We recall the following lemma for the sake of self-containedness:

Lemma B.2. For $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ Lipschitz continuous and $\mu, \nu \in \mathcal{P}(\mathbb{R}^n)$,

$$W_2(F_\# \mu, F_\# \nu) \leq L_F W_2(\mu, \nu) \quad (12)$$

Proof. Assume $\gamma^* \in \Gamma(\mu, \nu)$ is the optimal coupling between μ and ν , hence

$$\gamma^*(\cdot, \mathbb{R}^n) = \mu, \gamma^*(\mathbb{R}^n, \cdot) = \nu, \text{ and } \int_{\mathbb{R}^n \times \mathbb{R}^n} |\mathbf{x} - \mathbf{y}|^2 d\gamma^*(\mathbf{x}, \mathbf{y}) = W_2(\mu, \nu)^2. \quad (13)$$

Consider the coupling $\gamma_F := (F \times F)_\# \gamma^*$. Take $B \subset \mathbb{R}^m$ measureable. Then,

$$\begin{aligned} \gamma_F(B \times \mathbb{R}^m) &= ((F \times F)_\# \gamma^*)(B \times \mathbb{R}^m) = \gamma^*((F \times F)^{-1}(B \times \mathbb{R}^m)) \\ &= \gamma^*(F^{-1}(B) \times F^{-1}(\mathbb{R}^m)) = \gamma^*(F^{-1}(B) \times \mathbb{R}^n) = \mu(F^{-1}(B)) = (F_\# \mu)(B). \end{aligned} \quad (14)$$

Analogously, $\gamma_F^*(\mathbb{R}^m \times B) = (F_\# \nu)(B)$. Hence γ_F^* is a valid competitor for the optimal transport problem from $F_\# \mu$ to $F_\# \nu$:

$$\begin{aligned} W_2(F_\# \mu, F_\# \nu)^2 &= \min_{\gamma \in \Gamma(F_\# \mu, F_\# \nu)} \int_{\mathbb{R}^m \times \mathbb{R}^m} |\mathbf{x} - \mathbf{y}|^2 d\gamma(\mathbf{x}, \mathbf{y}) \leq \int_{\mathbb{R}^m \times \mathbb{R}^m} |\mathbf{x} - \mathbf{y}|^2 d\gamma_F(\mathbf{x}, \mathbf{y}) \\ &= \int_{\mathbb{R}^n \times \mathbb{R}^n} |F(\mathbf{x}) - F(\mathbf{y})|^2 d\gamma^*(\mathbf{x}, \mathbf{y}) \leq L_F^2 \int_{\mathbb{R}^n \times \mathbb{R}^n} |\mathbf{x} - \mathbf{y}|^2 d\gamma^*(\mathbf{x}, \mathbf{y}) = L_F^2 W_2(\mu, \nu)^2 \end{aligned} \quad (15)$$

Taking the square root gives the claimed result. $\square$

Proposition (3.1). Let F interpolate the training data D_ξ as in (4). At any time t , consider $\tilde{M}$ test samples $\mathcal{D}_\xi^{\text{test}} = \{(\tilde{\mathbf{x}}_t^i, \tilde{\mathbf{x}}_{t+1}^i, \tilde{\xi}_t^i)\}_{i=1}^{\tilde{M}}$ sampled from $\pi_t \otimes \nu$. Evaluate now the map F to obtain the generated samples $\hat{\mathbf{x}}_t^i = F(\tilde{\mathbf{x}}_t^i, \tilde{\xi}_t^i)$ for $i = 1, \dots, \tilde{M}$. Then,

$$\mathbb{E}_{\mathcal{D}_\xi, \mathcal{D}_\xi^{\text{test}}} [W_2(\hat{\rho}_{t+1}, \tilde{\rho}_{t+1})^2] \leq 2C(1 + L_F^2) \min(M, \tilde{M})^{-1/\alpha}, \quad (16)$$

where $\hat{\rho}_{t+1}$ and $\tilde{\rho}_{t+1}$ are the empirical measures corresponding to the generated samples $\{\hat{\mathbf{x}}_{t+1}^i\}_{i=1}^{\tilde{M}}$ and the test samples $\{\tilde{\mathbf{x}}_{t+1}^i\}_{i=1}^{\tilde{M}}$, $C > 0$ is a constant, L_F is the Lipschitz constant of F , and $\alpha \geq 0$ depends on the intrinsic dimension of ρ_t, ρ_{t+1} , or ν , whichever is largest. When $F(\mathbf{x}, \xi) = \mathbf{x} + dtR(\mathbf{x}, \xi)$, equation 5 can be written as

$$\mathbb{E}_{\mathcal{D}_\xi, \mathcal{D}_\xi^{\text{test}}} [W_2(\hat{\rho}_{t+1}, \tilde{\rho}_{t+1})^2] \leq C(3 + 4dt^2 L_R^2) \min(M, \tilde{M})^{-2/\alpha}. \quad (17)$$

If $F(\mathbf{x}, \xi) = \mathbf{x} + dt b(\mathbf{x}) + \varepsilon \sigma(\xi)$, then (5) can be written as

$$\mathbb{E}_{\mathcal{D}_\xi, \mathcal{D}_\xi^{\text{test}}} [W_2(\hat{\rho}_{t+1}, \tilde{\rho}_{t+1})^2] \leq C(3 + 4dt^2 L_b^2 + 4\varepsilon^2 L_\sigma^2) \min(M, \tilde{M})^{-2/\alpha}. \quad (18)$$

Notice that the strong concentration bound (Weed & Bach, 2019)[Proposition 20] for empirical measures $\hat{\mu}$ and μ imply that control over the expectation is sufficient to bound the error with high probability,

$$\mathbb{P} [W_2(\mu, \hat{\mu})^2 \geq \mathbb{E} [W_2(\mu, \hat{\mu})^2] + \varepsilon] \leq \exp(-2M\varepsilon^2), \quad \varepsilon > 0. \quad (19)$$

Proof. By the triangle inequality,

$$W_2(F_{\#}(\tilde{\rho}_t \otimes \tilde{\nu}), \tilde{\rho}_{t+1}) \leq \underbrace{W_2(F_{\#}(\hat{\rho}_t \otimes \hat{\nu}), \hat{\rho}_{t+1})}_{=: (i)} + \underbrace{W_2(\hat{\rho}_{t+1}, \tilde{\rho}_{t+1})}_{=: (ii)} + \underbrace{W_2(F_{\#}(\tilde{\rho}_t \otimes \tilde{\nu}), F_{\#}(\hat{\rho}_t \otimes \hat{\nu}))}_{=: (iii)} \quad (20)$$

The first term (i) is zero by the assumption that F interpolates the training data.

The second term (ii) is bounded by the Wasserstein distance between the training and test data. Its expectation value can be controlled by the fact that both empirical measures are drawn from the same distribution. In (Weed & Bach, 2019)[Proposition 5], it is shown that for an empirical measures $\hat{\mu} \approx \mu$ with M samples,

$$\mathbb{E} [W_2(\mu, \hat{\mu})^2] \leq CM^{-2/\alpha}, \quad (21)$$

where α denotes the intrinsic dimension of the data. Hence

$$\mathbb{E} [W_2(\hat{\rho}_{t+1}, \tilde{\rho}_{t+1})^2] \leq C_{t+1}(M^{-2/\alpha_{t+1}} + \tilde{M}^{-2/\alpha_{t+1}}). \quad (22)$$

For the third term (iii), we can estimate

$$\mathbb{E} [W_2(F_{\#}(\tilde{\rho}_t \otimes \tilde{\nu}), F_{\#}(\hat{\rho}_t \otimes \hat{\nu}))] \leq \mathbb{E} [L_F W_2(\tilde{\rho}_t \otimes \tilde{\nu}, \hat{\rho}_t \otimes \hat{\nu})] \quad (23)$$

where L_F is the Lipschitz constant of F :

$$|F(\mathbf{x}, \boldsymbol{\xi}) - F(\mathbf{x}', \boldsymbol{\xi}')| \leq L_F |(\mathbf{x}, \boldsymbol{\xi}) - (\mathbf{x}', \boldsymbol{\xi}')| \quad \forall (\mathbf{x}, \boldsymbol{\xi}, \mathbf{x}', \boldsymbol{\xi}') \in \mathcal{X} \times \mathbb{R}^d \times \mathcal{X} \times \mathbb{R}^d \quad (24)$$

Now, note that the quadratic Wasserstein distance in this case factorizes as

$$W_2(\tilde{\rho}_t \otimes \tilde{\nu}, \hat{\rho}_t \otimes \hat{\nu})^2 = W_2(\tilde{\rho}_t, \hat{\rho}_t)^2 + W_2(\tilde{\nu}, \hat{\nu})^2 \quad (25)$$

and hence

$$\mathbb{E} [W_2(\tilde{\rho}_t \otimes \tilde{\nu}, \hat{\rho}_t \otimes \hat{\nu})^2] \leq C_t(M^{-2/\alpha_t} + \tilde{M}^{-2/\alpha_t}) + C_{\nu}(M^{-2/\alpha_{\nu}} + \tilde{M}^{-2/\alpha_{\nu}}). \quad (26)$$

Collecting all terms, we get

$$\begin{aligned} & \mathbb{E} [W_2(F_{\#}(\tilde{\rho}_t \otimes \tilde{\nu}), \tilde{\rho}_{t+1})^2] \\ & \leq C_{t+1}(M^{-2/\alpha_{t+1}} + \tilde{M}^{-2/\alpha_{t+1}}) + L_F^2 C_t(M^{-2/\alpha_t} + \tilde{M}^{-2/\alpha_t}) + L_F^2 C_{\nu}(M^{-2/\alpha_{\nu}} + \tilde{M}^{-2/\alpha_{\nu}}) \end{aligned} \quad (27)$$

With $\alpha = \max(\alpha_t, \alpha_{t+1}, \alpha_{\nu})$ and $C = \max(C_t, C_{t+1}, C_{\nu})$, we obtain the result from Proposition 3.1.

To show the corollary, we use that the optimal coupling between $\tilde{\rho}_t \otimes \tilde{\nu}$ and $\hat{\rho}_t \otimes \hat{\nu}$ is given by $\gamma_x \otimes \gamma_{\xi}$, where γ_x is the optimal coupling between $\tilde{\rho}_t$ and $\hat{\rho}_t$ and γ_{ξ} is the optimal coupling between $\tilde{\nu}$ and $\hat{\nu}$:

$$W_2(F_{\#}(\tilde{\rho}_t \otimes \tilde{\nu}), F_{\#}(\hat{\rho}_t \otimes \hat{\nu}))^2 \leq \int_{\mathcal{X}^2 \times \mathbb{R}^{2d}} |F(\hat{\mathbf{x}}, \hat{\boldsymbol{\xi}}) - F(\tilde{\mathbf{x}}, \tilde{\boldsymbol{\xi}})|^2 d\gamma_x \otimes \gamma_{\xi} \quad (28)$$

$$= \int_{\mathcal{X}^2 \times \mathbb{R}^{2d}} |\hat{\mathbf{x}} - \tilde{\mathbf{x}} + dt(R(\hat{\mathbf{x}}, \hat{\boldsymbol{\xi}}) - R(\tilde{\mathbf{x}}, \tilde{\boldsymbol{\xi}}))|^2 d\gamma_x \otimes \gamma_{\xi} \quad (29)$$

Next, Young's inequality implies that

$$2|\hat{\mathbf{x}} - \tilde{\mathbf{x}}| |R(\hat{\mathbf{x}}, \hat{\boldsymbol{\xi}}) - R(\tilde{\mathbf{x}}, \tilde{\boldsymbol{\xi}})| \leq \frac{1}{\varepsilon} |\hat{\mathbf{x}} - \tilde{\mathbf{x}}|^2 + \varepsilon |R(\hat{\mathbf{x}}, \hat{\boldsymbol{\xi}}) - R(\tilde{\mathbf{x}}, \tilde{\boldsymbol{\xi}})|^2 \quad (30)$$

for any $\varepsilon > 0$ and hence, with $\varepsilon = dt$,

$$|\hat{\mathbf{x}} - \tilde{\mathbf{x}} + dt(R(\hat{\mathbf{x}}, \hat{\boldsymbol{\xi}}) - R(\tilde{\mathbf{x}}, \tilde{\boldsymbol{\xi}}))|^2 \leq 2 \left(|\hat{\mathbf{x}} - \tilde{\mathbf{x}}|^2 + dt^2 |R(\hat{\mathbf{x}}, \hat{\boldsymbol{\xi}}) - R(\tilde{\mathbf{x}}, \tilde{\boldsymbol{\xi}})|^2 \right) \quad (31)$$

and

$$(29) \leq 2 \int_{\mathcal{X} \times \mathcal{X}} |\hat{\mathbf{x}} - \tilde{\mathbf{x}}|^2 d\gamma_x + 2dt^2 \int_{\mathcal{X}^2 \times \mathbb{R}^{2d}} |R(\hat{\mathbf{x}}, \hat{\boldsymbol{\xi}}) - R(\tilde{\mathbf{x}}, \tilde{\boldsymbol{\xi}})|^2 d\gamma_x \otimes \gamma_\xi \quad (32)$$

$$\leq 2W_2(\tilde{\rho}_t, \hat{\rho}_t)^2 + 2dt^2 L_R^2 W_2(\tilde{\rho}_t \otimes \nu, \hat{\rho}_t \otimes \nu)^2 \quad (33)$$

$$= 2W_2(\tilde{\rho}_t, \hat{\rho}_t)^2 + 2dt^2 L_R^2 (W_2(\tilde{\rho}_t, \hat{\rho}_t)^2 + W_2(\tilde{\nu}, \hat{\nu})^2) \quad (34)$$

The proof for the case $F(\mathbf{x}, \boldsymbol{\xi}) = \mathbf{x} + dt b(\mathbf{x}) + \varepsilon \sigma(\boldsymbol{\xi})$ follows analogously, using

$$|\hat{\mathbf{x}} - \tilde{\mathbf{x}} + dt(b(\hat{\mathbf{x}}) - b(\tilde{\mathbf{x}})) + \varepsilon(\sigma(\hat{\boldsymbol{\xi}})) - \sigma(\tilde{\boldsymbol{\xi}})|^2 \leq 2|\hat{\mathbf{x}} - \tilde{\mathbf{x}}|^2 + 2dt^2 |b(\hat{\mathbf{x}}) - b(\tilde{\mathbf{x}})|^2 + \frac{\varepsilon^2}{dt^2} |\sigma(\hat{\boldsymbol{\xi}}) - \sigma(\tilde{\boldsymbol{\xi}})|^2. \quad (35)$$

□

C. Explicit construction of an interpolating map

Assume $\boldsymbol{\xi}_t$ is distributed uniformly on the unit sphere $\mathbb{S}^{d-1}$ and $d \geq M$. Consider the following general form for F :

$$F(\mathbf{x}_t, \boldsymbol{\xi}_t) = b(\mathbf{x}_t) + \sum_{i,k=1}^M (\hat{\mathbf{x}}_{t+1}^i - b(\mathbf{x}_t)) \hat{\mathbb{G}}_{ik}^{-1} \hat{\boldsymbol{\xi}}_t^k \cdot \boldsymbol{\xi}_t, \quad (36)$$

where $b : \mathcal{X} \rightarrow \mathcal{X}$ captures the mean behavior, i.e. $b(\mathbf{x}_t) = \mathbb{E}[\mathbf{X}_{t+1} | \mathbf{X}_t = \mathbf{x}_t]$, $(\hat{\mathbf{x}}_{t+1}^i, \hat{\boldsymbol{\xi}}_t^i)$, $i = 1, \dots, M$ are the training points, and $\hat{\mathbb{G}}_{ik} = \hat{\boldsymbol{\xi}}_t^i \cdot \hat{\boldsymbol{\xi}}_t^k$. Note that $\hat{\mathbb{G}}$ is almost surely invertible when $d \geq M$.

This function interpolates the training data:

$$F(\hat{\mathbf{x}}_t^j, \hat{\boldsymbol{\xi}}_t^j) = b(\mathbf{x}_t) + \sum_{i,k=1}^M (\hat{\mathbf{x}}_{t+1}^i - b(\mathbf{x}_t)) \hat{\mathbb{G}}_{ik}^{-1} \hat{\boldsymbol{\xi}}_t^k \cdot \boldsymbol{\xi}_t^j \quad (37)$$

$$= b(\mathbf{x}_t) + \sum_{i=1}^M (\hat{\mathbf{x}}_{t+1}^i - b(\mathbf{x}_t)) \delta_{ij} = \hat{\mathbf{x}}_{t+1}^j \quad \forall j = 1, \dots, M \quad (38)$$

At the same time, it is an affine function in $\boldsymbol{\xi}_t$ and as regular in $\mathbf{x}_t$ as b is, and the latter is smooth in physical systems and natural videos. We can improve the regularity of F by improving the conditioning of $\hat{\mathbb{G}}$. As discussed, $\hat{\mathbb{G}}$ approaches the identity matrix for large d and fixed M .

Furthermore, note that for any $\mathbf{v} \in \mathbb{S}^{d-1}$, $\sqrt{d} \mathbf{v} \cdot \boldsymbol{\xi}_t \rightarrow \eta \sim \mathcal{N}(0, 1)$ in distribution as $d \rightarrow \infty$ (Vershynin, 2018)[Theorem 3.3.9]. For fixed $\mathbf{x}_t$, the distribution of $F(\mathbf{x}_t, \boldsymbol{\xi}_t)$ is thus similar to

$$b(\mathbf{x}_t) + \frac{1}{\sqrt{d}} \sum_{i=1}^M (\hat{\mathbf{x}}_{t+1}^i - b(\mathbf{x}_t)) \eta_i \quad \text{where} \quad \eta_i \sim \mathcal{N}(0, 1) \quad \forall i, \quad (39)$$

namely a Gaussian perturbation of the expected value of $\mathbf{X}_{t+1} | \mathbf{X}_t = \mathbf{x}_t$ in the directions of other training points.

D. Nearest Neighbor analysis

We conduct a nearest neighbor analysis to ensure that our model generates novel samples rather than memorizing training data. Specifically, for each example: the top row shows a randomly selected generated sample, the middle row displays the closest matching sample from the training set, and the bottom row shows the closest training-set neighbor to the middle row itself. The visual variation between the generated sample (top row) and its nearest training-set neighbor (middle row) is similar in scale to the variation observed between the two training-set samples (middle and bottom rows). This demonstrates that our model generalizes effectively without simply replicating training examples.

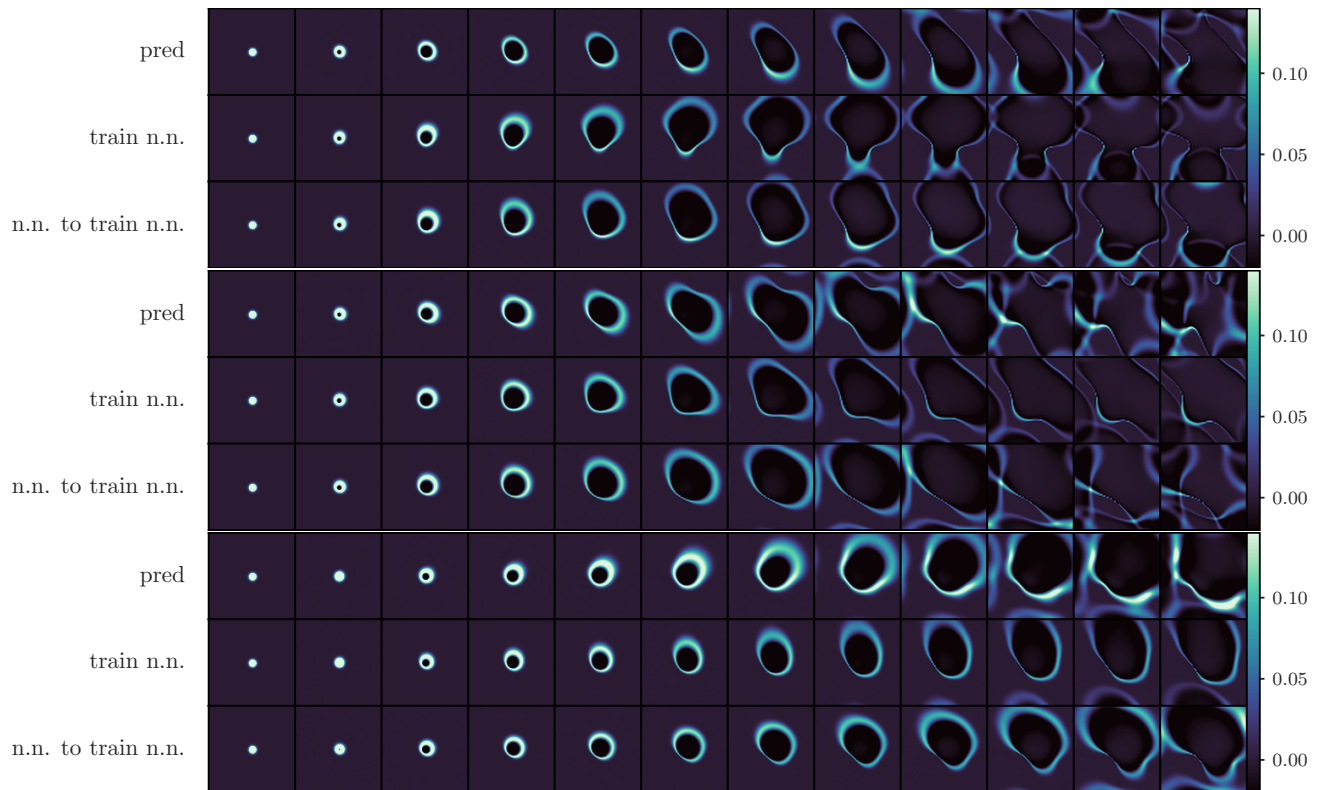


Figure 7. Wave nearest neighbor analysis. Of each image: top row randomly chosen generated sample, middle row nearest neighbor in the training set, bottom row nearest neighbor in the training set to the middle row. The variation between the top and middle row should be roughly similar to the variation between the middle and bottom row, indicating generalization without memorization.

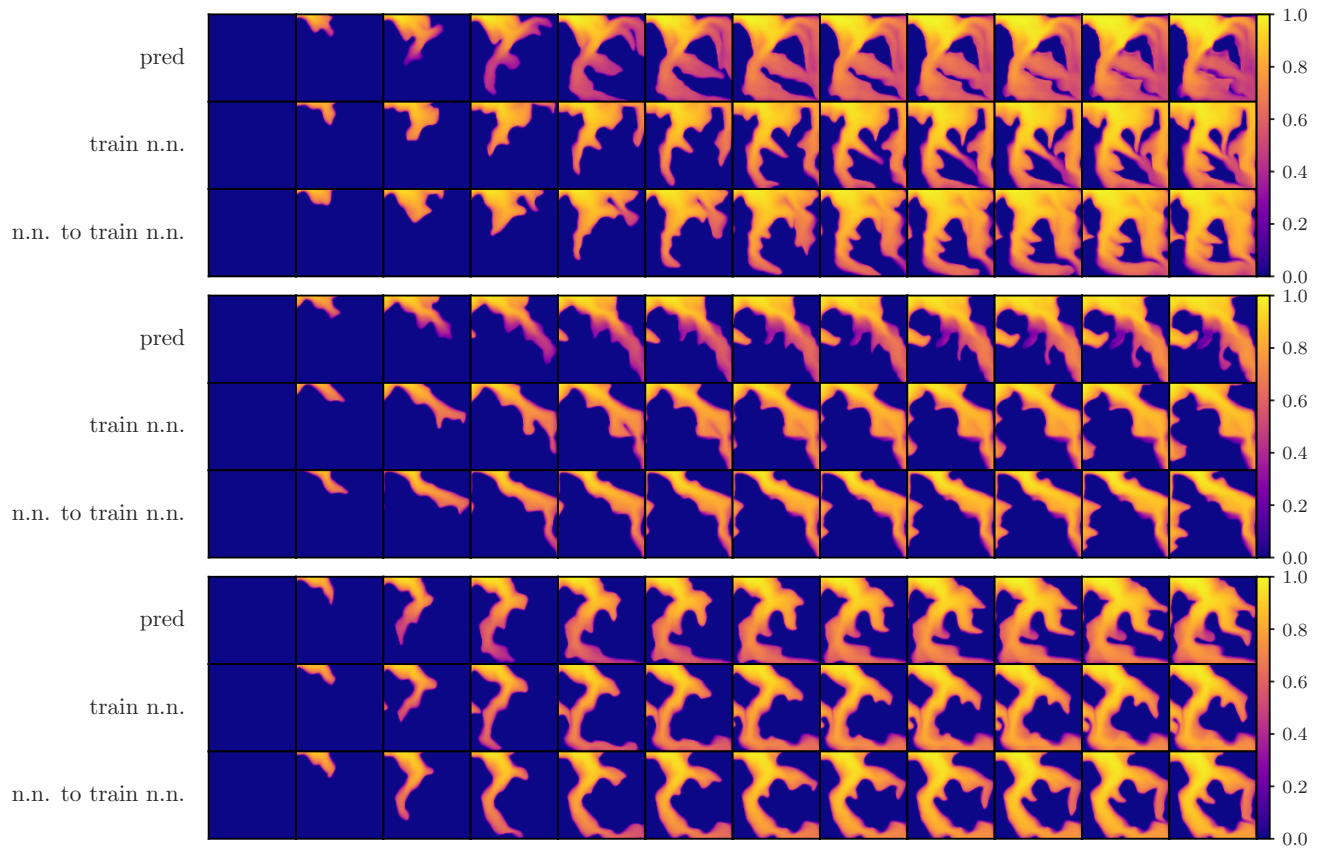


Figure 8. Flow nearest neighbor analysis. Of each image: top row randomly chosen generated sample, middle row nearest neighbor in the training set, bottom row nearest neighbor in the training set to the middle row. The variation between the top and middle row should be roughly similar to the variation between the middle and bottom row, indicating generalization without memorization.

E. Numerical experiments

E.1. Duffing SDE

The physical trajectories $x_t \in \mathbb{R}^2$ follow the SDE with

$$\frac{d}{dt} \begin{bmatrix} x^1 \\ x^2 \end{bmatrix} (t) = \begin{bmatrix} x^2 \\ -0.4x^2 + 0.04x^1 + 0.008(x^1)^3 \end{bmatrix} (t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \xi(t). \quad (40)$$

The initial configuration is a Gaussian centered at $m_0 = [0, 10]$ with diagonal variance of strength 0.5. The dynamics are integrated over $0 \leq t \leq 14$ for $N = 5000$ samples.

E.2. Traveling wave through random media (Wave)

The initial condition is a Gaussian bump at the center of the domain. As the wave evolves, the random medium leads to different wave speeds that lead to a deformation of the wave front, which leads to a distribution of diverse solution fields; see Appendix H.1 for visualizations. Importantly, for both wave and flow problem, the initial condition is deterministic and fixed, thus deterministic approaches such as neural operators cannot capture the stochasticity induced by the random media and permeability fields during the autoregressive roll out; see Section 2.2.

Problem setup:

$$\begin{cases} \partial_{tt}u(t, \mathbf{x}) = c^2(\mathbf{x})\Delta u(t, \mathbf{x}), & t \in (0, 8], \mathbf{x} \in \Omega, \\ u(0, \mathbf{x}) = \exp(-30 \|\mathbf{x} - [\pi, \pi]\|^2), & \mathbf{x} \in \Omega, \\ \partial_t u(0, \mathbf{x}) = 0, & \mathbf{x} \in \Omega. \end{cases}$$

$\Omega = [0, 2\pi] \times [0, 2\pi]$, $T = 8.0$ We look at the 2D wave equation with periodic boundary conditions.

The wave speed $c(\mathbf{x})$ varies in space; it is generated by choosing random Fourier modes from a log-normal distribution which peaks at some wave number $k = 1$ which controls the regularity of the random field.

We have $M = 1024$ trajectories, discretized at $T = 64$ time points and 64×64 points in space.

E.3. Two-phase flow in random porous media (Flow)

We consider an incompressible and immiscible two-phase flow system of water and oil phases (Aarnes et al., 2007). One phase of the flow moves through the domain as it interacts with a randomly permeability field which deforms the flow into a high variance distribution of complex shapes; see appendix H.2. The governing equations are

$$\begin{aligned} -\nabla \cdot (K\lambda(s)\nabla p) &= q, \\ \phi \frac{\partial}{\partial t} s + \nabla \cdot (f(s)v) &= \frac{q_\omega}{\rho_\omega}, \end{aligned}$$

where s is the saturation field, K represents the permeability tensor, $\lambda(s)$ is the total fluid mobility (the sum of water and oil mobilities), and q is the source term. In the second equation, ϕ denotes the porosity, $f(s)$ the fractional flow of water, and v is the Darcy velocity.

The randomness enters via the permeability field K , is the exponential $K(x) = \exp(\epsilon Z(x) + \bar{Z} + Y(x))$, where Z and Y are independent Gaussian random fields, each generated with a Matérn covariance function (marginal standard deviation $\sigma = 3.0$, correlation length $\lambda = 10$, smoothness parameter $\nu = 1$, noisiness parameter $\epsilon = 0.01$).

The equations are discretized with an explicit upwind finite-volume discretization method. We use 100×100 cells, end time 0.7, and time-step size 0.007. The initial condition is zero saturation with a localized source on the upper left corner of the spatial domain.

E.4. Video data sets

We measure inference runtime by using the largest batch size that fits into the memory of our H100 GPUs and then divide the total runtime by that batch to get the per-video generation runtime.

E.5. Metrics

For our physics-based examples, we regard the realizations of our stochastic process as time-dependent solution fields. That is, the j -th component of a sample $\mathbf{x}_t$ corresponds to the evaluation of $u : \mathcal{T} \times [0, L]^2 \rightarrow \mathbb{R}$ at the j -th grid point in $[0, L]^2$ and at time t .

To ease comparisons, we normalize $\mathcal{T}$ to $[0, 1]$ and L to 1 when computing all metrics.

For the sake of comparison and visualization, we require low-dimensional metrics. We construct them as follows: Take a map $f : \mathbf{x}_t \mapsto f(\mathbf{x}_t) \in \mathbb{R}$. The push-forward of the empirical distribution of $\mathbf{x}_t$ under f defines a distribution on $\mathbb{R}$.

Given samples from the ground truth $\mathbf{x}_t \sim \hat{\rho}_t$ and generated samples $\tilde{\mathbf{x}}_t \sim \tilde{\rho}_t$, the mismatch can be described by $W_2(f_{\#}\hat{\rho}_t, f_{\#}\tilde{\rho}_t)$, which is easy to compute as these are one-dimensional distributions.

The mismatch in this case is defined as a function of time t . It is also possible to consider a map g that takes $\{\mathbf{x}_t\}_t$ an entire trajectory and returns a one-dimensional quantity of interest.

Wasserstein distance of Mass We look at Mass Wasserstein distance (WDM). In this case, the function f corresponds to an l_1 norm, denoted m :

$$m(\mathbf{x}_t) = \frac{1}{n} \|\mathbf{x}_t\|_1 \approx \int_{[0, L]^2} |u(t, x)| dx, \quad \text{MWD} = \frac{1}{T} \sum_{t=1}^T W_2(m_{\#}\hat{\rho}_t, m_{\#}\tilde{\rho}_t).$$

Wasserstein crossing time Given some subset $B \subset [0, L]^2$ the earliest arrival time τ for some threshold value $c \in \mathbb{R}$ is given by

$$\tau(u) = \min\{t \in \mathcal{T} : u(t, x) > c \text{ for } x \in B\}.$$

For a trajectory $\{\mathbf{x}_t\}_t$, this corresponds to a threshold value being crossed for a subset of entries of $\mathbf{x}_t$. Denote by ρ the union of the time marginals ρ_t for all t . The Wasserstein crossing time is defined as

$$\text{WCT} = W_2(\tau_{\#}\hat{\rho}, \tau_{\#}\tilde{\rho}).$$

F. Architecture details

UNet backbone Our network follows the canonical encoder–decoder UNet with skip–connections, implemented in `Flax`. All architecture decisions are standard. We use GroupNorm, no dropout, stride-2 3×3 convolution. At each spatial scale for our residual blocks the channel depths are,

$$\begin{aligned} \text{medium_feature_depths} &= [128, 256, 512], \\ \text{large_feature_depths} &= [128, 256, 512, 1024]. \end{aligned}$$

The only architectural modification is the conditioning on the label ξ_t . Many diffusion backbones condition on the “diffusion time” via some modulation scheme: The only difference is we skip the initial sinusoidal embedding which maps the diffusion-time scalar to a vector. Instead we directly embed ξ with a two-layer MLP with width 512. The MLP output then modulates the residual feature maps via a standard FiLM modulation scheme (Perez et al., 2018).

As stated in the main text, apart from re-purposing the timestep embedding to encode the conditioning label ξ , all components adhere to established UNet practice, enabling a like-for-like comparison with standard diffusion backbones.

G. Training details

For the physics-based problems (wave, flow), we use the L2 loss defined in equation 8. For the video datasets, we replace the L2 metric ($\|x - y\|_2^2$) with a perceptual based LPIPS loss (Zhang et al., 2018) which we find improves FVD scores. This normalization prevents the network’s predictions from diverging during rollout, making it particularly suitable for video data, which naturally lies within $[0, 1]$.

Table 3. Hyper-parameters used for each dataset.

	Wave	Flow	BAIR	Clevrer
Batch Size	64	64	128	32
Label Dim	64	64	512	256
Iterations	350 000	350 000	500 000	500 000
Optimizer	AdamW	AdamW	AdamW	AdamW
Learning Rate	$1e^{-4}$	$1e^{-4}$	$1e^{-4}$	$1e^{-4}$
Schedule	cosine	cosine	cosine	cosine
Network Size	Medium	Medium	Medium	Medium
Loss	L2	L2	LPIPS	LPIPS
Trajectories (M)	1024	1000	43 000	10 000
Time Points (T)	64	50	16	16
State Dim (n)	64^2	96^2	$64^2 \times 3$	$128^2 \times 3$
Markov Window (m)	3	3	1	3

H. Uncurated Samples

H.1. Wave

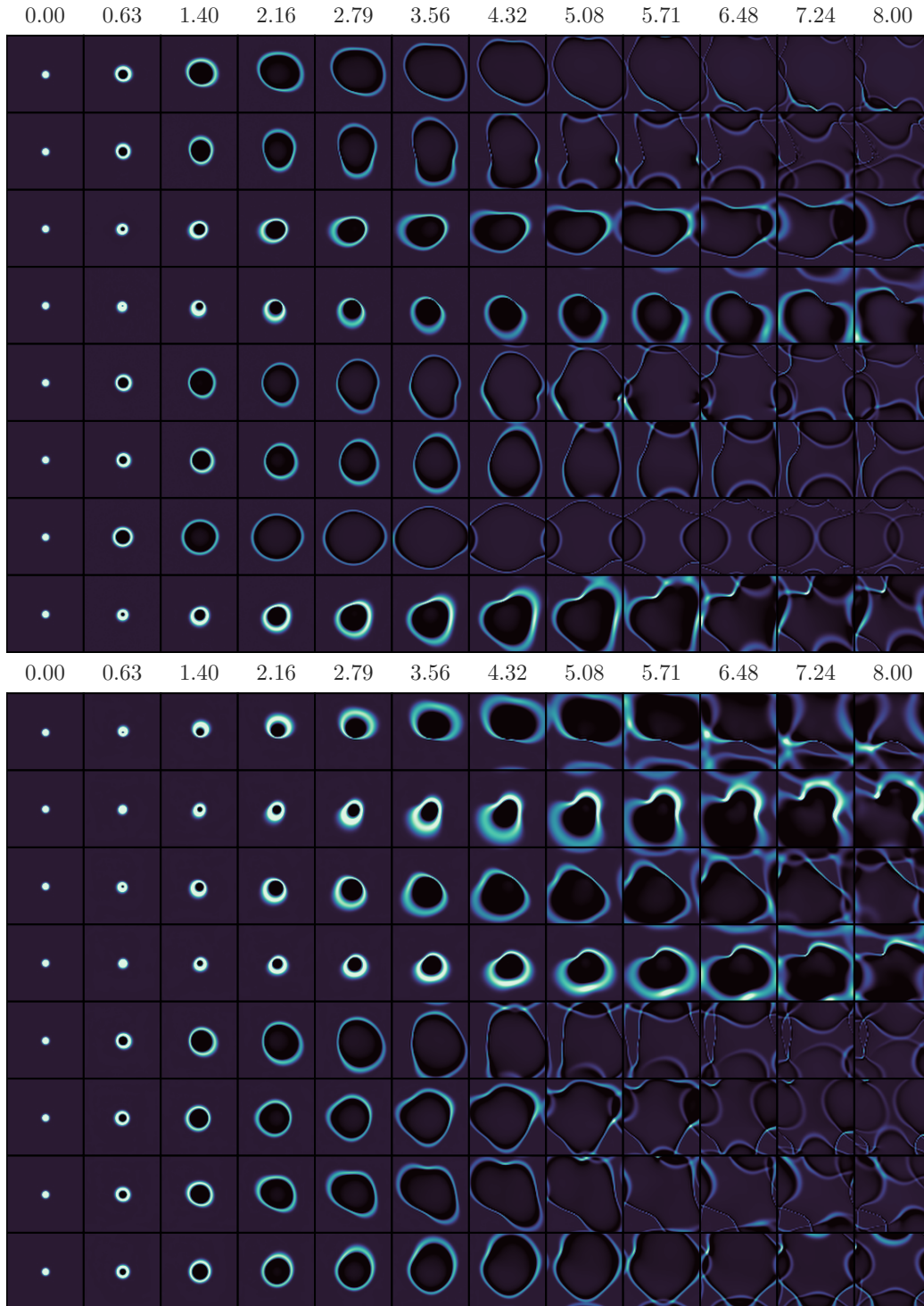


Figure 9. Wave samples (uncurated). Top: true samples. Bottom: generated from Stochastic Lifting.

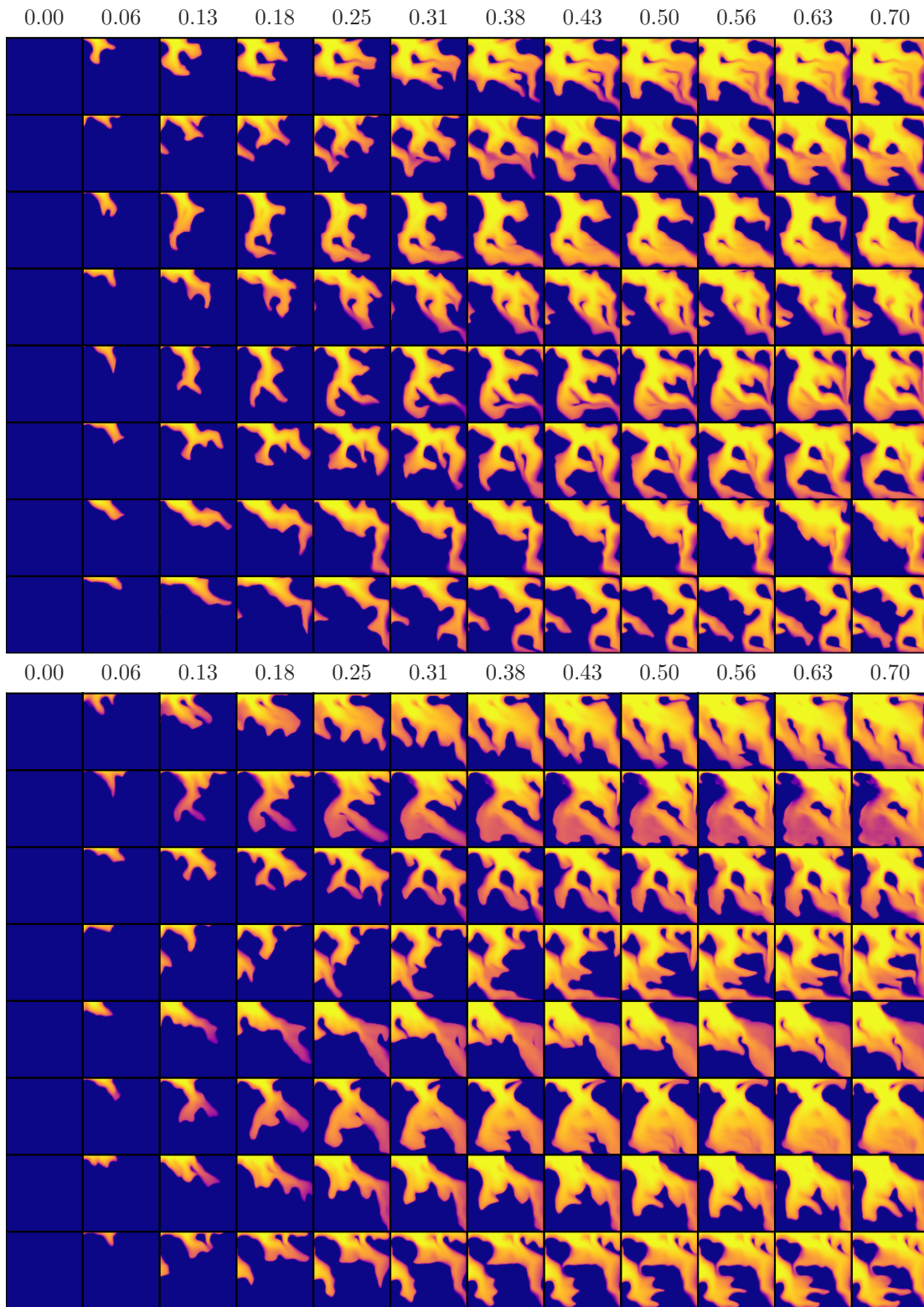
H.2. Flow

Figure 10. **Flow samples (uncurated).** **Top:** true samples. **Bottom:** generated from Stochastic Lifting.

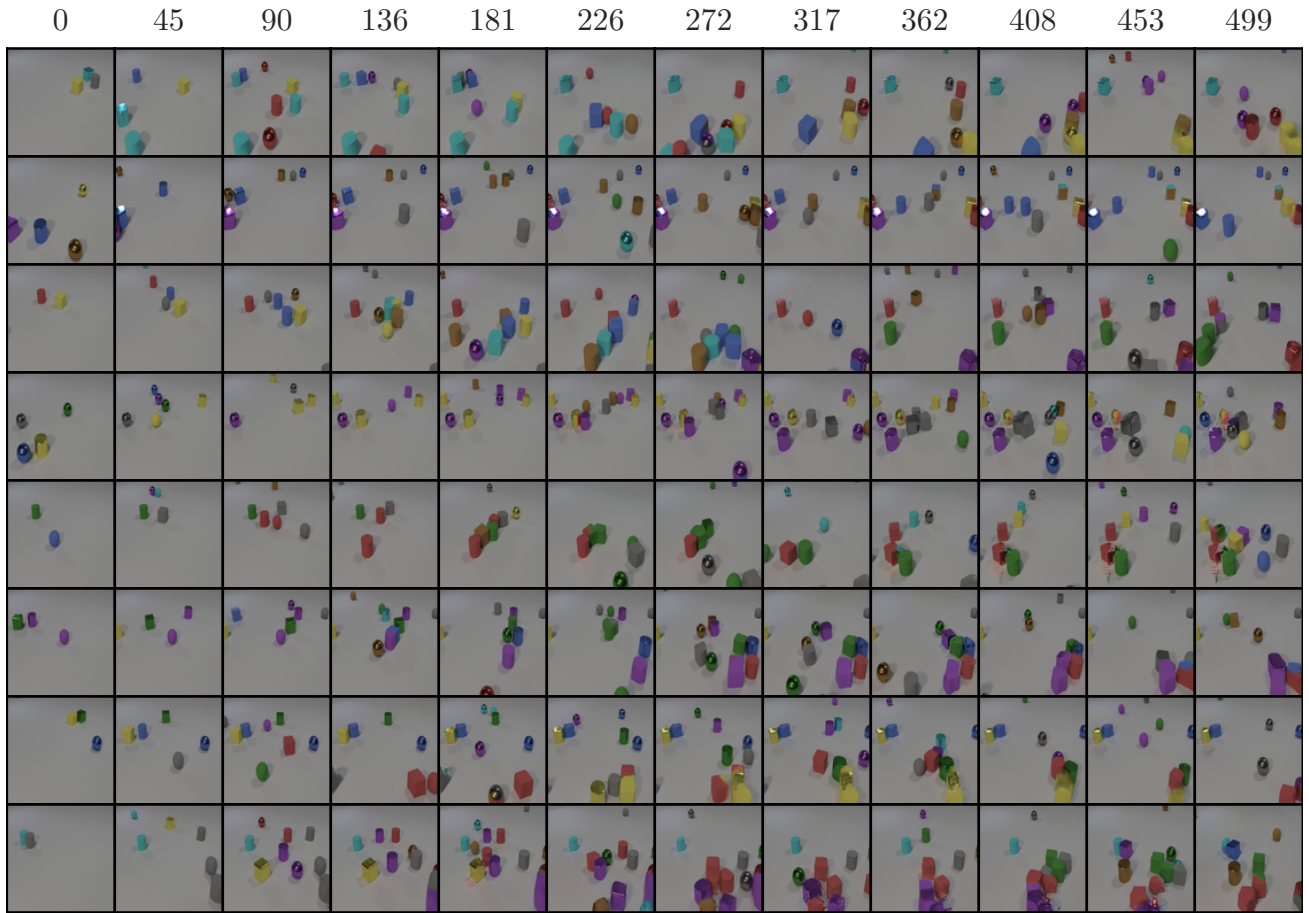
H.3. CLEVRER

Figure 11. **CLEVRER** samples (uncurated). Generated from Stochastic Lifting. Long rollout.

I. Further theoretical arguments

In the following, assume that $\mathbf{x}_t^i \equiv \mathbf{x}_t$ for all i and we omit dependence on it from now on. (this is the one-to-many situation)

I.1. Adding reordering to the SL loss

For the sake of simplicity, assume that the label dimension d is equal to the physical dimension n . We can modify the stochastic lifting loss as follows: We denote with $\hat{\nu}_t$ the empirical distribution of the labels $\{\xi_t^i\}_{i=1}^M$ and with $\hat{\rho}_{t+1}$ that of the states $\{\mathbf{x}_{t+1}^i\}_{i=1}^M$. By computing $W_2^2(\hat{\nu}_t, \hat{\rho}_{t+1})$ for all $t = 0, \dots, T-1$, a structured coupling π_t is found. In particular, the coupling π_t defines a permutation σ_t such that $\mathbf{x}_{t+1}^i$ is coupled with (i.e. transported to) $\xi_t^{\sigma_t(i)}$.

With this information, the SL loss can be modified to

$$\mathcal{L}(\theta, \mathcal{D}_\xi) = \frac{1}{MT} \sum_{i=1}^M \sum_{t=0}^{T-1} \|F_\theta(\mathbf{x}_t^i, \xi_t^{\sigma_t(i)}) - \mathbf{x}_{t+1}^i\|_2^2. \quad (41)$$

Training not on ξ_t^i (random coupling) but on $\xi_t^{\sigma_t(i)}$ guarantees that the function F minimizing the modified SL loss is structured in a way such that the known regularity results for optimal transport maps apply.

$$W_2(\rho(|x_i|), \rho(|x_j|)) \leq L\|x_i - x_j\|$$

I.2. Finite samples in high dimensions

When the number of samples is of the same order as the dimension d (here assumed equal to n for simplicity), there is in fact little difference in the transport cost $\sum_{ij} c_{ij} \pi_{ij}$ of a random coupling and the W_2 -optimal coupling π^* . Heuristically, this is true because we can find a coordinate realization in which all samples are nearly orthogonal, $|\xi_t^i \cdot \mathbf{x}_{t+1}^j| < \frac{\epsilon}{4} \forall i, j$. Then, using $\sum_i \pi_{ij} = \sum_j \pi_{ij} = \frac{1}{M}$ and $\pi_{ij} \geq 0$ for any admissible plan,

$$\sum_{ij} |\xi_t^i - \mathbf{x}_{t+1}^j|^2 \pi_{ij} = \sum_{ij} (|\xi_t^i|^2 - 2\xi_t^i \cdot \mathbf{x}_{t+1}^j + |\mathbf{x}_{t+1}^j|^2) \pi_{ij} \quad (42)$$

$$= \frac{1}{M} \left(\sum_i |\xi_t^i|^2 + \sum_j |\mathbf{x}_{t+1}^j|^2 \right) - 2 \sum_{ij} \xi_t^i \cdot \mathbf{x}_{t+1}^j \pi_{ij}. \quad (43)$$

Since $|2 \sum_{ij} \xi_t^i \cdot \mathbf{x}_{t+1}^j \pi_{ij}| \leq \frac{\epsilon}{2} \sum_{ij} |\pi_{ij}|$, we see that all admissible plans (including the W_2 -optimal one) differ in their associated cost by at most

$$\left| 2 \sum_{ij} \xi_t^i \cdot \mathbf{x}_{t+1}^j (\pi_{ij} - \pi_{ij}^*) \right| \leq \frac{\epsilon}{2} \sum_{ij} (\pi_{ij} + \pi_{ij}^*) = \epsilon. \quad (44)$$

I.3. Concentration of measure in high dimensions

In high dimensions and on "shell-like" distributions (such as log-concave or Gaussian data), the distance between two random samples concentrates tightly around the average a . The argument is similar as before: On a sphere of radius R in $\mathbb{R}^d$, it holds that $\mathbb{E}[\xi_t^i \cdot \mathbf{x}_{t+1}^j] = 0$ and $\text{Var}[\xi_t^i \cdot \mathbf{x}_{t+1}^j] = d^{-1} R^4$. Using Chebychev's inequality, we can derive the concentration bound

$$P(|\xi_t^i \cdot \mathbf{x}_{t+1}^j| > \epsilon) \leq \frac{R^4}{d\epsilon^2}. \quad (45)$$

The average distance of two samples is $2R^2$ and the deviation from this value satisfies

$$P(|\xi_t^i - \mathbf{x}_{t+1}^j|^2 - 2R^2| > \epsilon) \leq \frac{4R^4}{d\epsilon^2}. \quad (46)$$

As a result, with probability $1 - \delta$, the deviation from $2R^2$ for a single sample pair is at most $\Delta = \frac{2R^2}{\sqrt{d\delta}}$. A very loose union bound then tells us that for all M^2 possible pairings, these are further than Δ away from $2R^2$ with probability at most $M^2\delta$.

Hence, with probability $1 - \delta'$, the deviation from $2R^2$ for all possible sample pairs is at most $\Delta' = \frac{2R^2 M}{\sqrt{d\delta'}}$.

The optimal transport cost is obtained by summing over all samples using the optimal permutation. This can be bounded above and below for every sample using its best and worst case pair,

$$c_{ij}(\pi_{ij} - \pi_{ij}^*) < 2\Delta' \quad (47)$$

and

$$|\sum_{ij} c_{ij}(\pi_{ij} - \pi_{ij}^*)| \leq M \cdot 2\Delta' = \frac{4R^2 M^2}{\sqrt{d\delta'}} \quad (48)$$

with probability $1 - \delta'$.

I.4. When does similar W_2 cost imply similar transport maps?

In general, two couplings π, π^* can satisfy $\sum_{ij} c_{ij}(\pi_{ij} - \pi_{ij}^*) < \epsilon$, yet $\sum_{ij} |\pi_{ij} - \pi_{ij}^*| \gg \epsilon$. Consider the case where two Dirac masses in $\mathbb{R}^2$ at $(-1, 0)$ and $(0, 1)$ are transported to two Dirac masses at $(\epsilon, 1)$ and $(-\epsilon, -1)$. Without mass splitting, there are two admissible transport plans, namely

$$\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad (49)$$

One has associated cost $1 + (1 + \epsilon)^2$ and the other $1 + (1 - \epsilon)^2$, hence their transport costs differ by 4ϵ , yet $\sum_{ij} |\pi_{ij} - \pi_{ij}^*| = 2$.

I.5. Relation between closeness of maps and closeness in cost

We have that

$$\|F - F^*\|_{L^2(\nu)}^2 = \|F - \text{id}\|_{L^2(\nu)}^2 + \|F^* - \text{id}\|_{L^2(\nu)}^2 - 2(F - \text{id}, F^* - \text{id})_{L^2(\nu)} \quad (50)$$

and hence

$$\|F - \text{id}\|_{L^2(\nu)}^2 - \|F^* - \text{id}\|_{L^2(\nu)}^2 = \|F - F^*\|_{L^2(\nu)}^2 + 2(F - F^*, F^* - \text{id})_{L^2(\nu)}. \quad (51)$$

If we can control the last term here, we can show that a small gap in cost (i.e. the transport cost of F is close to that of F^*) implies closeness of F and F^* .

Assume that F^* is the optimal transport map and its potential is strongly convex with constant λ . Then,

$$(F^*(x) - F^*(x')) \cdot (x - x') \geq \lambda |F^*(x) - F^*(x')|^2. \quad (52)$$

and it is known that

$$2(F - F^*, F^* - \text{id})_{L^2(\nu)} = 2(F - F^*, F^*) - 2(F - F^*, \text{id})_{L^2(\nu)} \geq -\frac{2}{\lambda} \|F - F^*\|_{L^2(\nu)}^2 \quad (53)$$

Hence,

$$\|F - F^*\|_{L^2(\nu)}^2 = \|F - \text{id}\|_{L^2(\nu)}^2 - \|F^* - \text{id}\|_{L^2(\nu)}^2 - 2(F - F^*, F^* - \text{id})_{L^2(\nu)} \quad (54)$$

and, together,

$$\left(1 - \frac{2}{\lambda}\right) \|F - F^*\|_{L^2(\nu)}^2 \leq \|F - \text{id}\|_{L^2(\nu)}^2 - \|F^* - \text{id}\|_{L^2(\nu)}^2. \quad (55)$$